

Investigating Blind Multichannel Identification Algorithms and Their Application for Speech Dereverberation Using MINT^{*}

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Abstract. Blind multichannel identification (BMCI) provides an unsupervised approach for estimating room impulse responses (RIRs) in reverberant acoustic scenarios, but its effectiveness for inverse filtering-based speech dereverberation remains insufficiently understood. In particular, conventional BMCI algorithms require a preset filter length, which is difficult to determine in real environments, making them inapplicable for subsequent dereverberation. This paper presents a unified evaluation of representative BMCI algorithms with respect to acoustic channel estimation accuracy and their applicability to MINT-based speech dereverberation. We compare solution-based BMCI methods with NMCFLMS-class adaptive algorithms under simulated and measured acoustic conditions, and analyze their robustness to filter-length mismatch. Results show that solution-based methods achieve accurate estimates only when the assumed length matches the true impulse response length, but degrade severely under mismatch. In contrast, NMCFLMS-class algorithms can stably provide RIR estimates as useful priors for MINT inverse filtering. By integrating these estimates with MINT, the proposed BMCI-MINT pipeline achieves higher PESQ scores than weighted prediction error (WPE) and generalized WPE baselines, even in the low-SNR scenarios. These findings demonstrate the potential of adaptive BMCI algorithms for practical multichannel dereverberation.

Keywords: blind multichannel identification · multichannel RIR estimation · multiple-input/output inverse theorem · speech dereverberation

1 Introduction

Blind multichannel identification (BMCI) aims to estimate unknown system impulse responses from multichannel observations without access to the source

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signal. Over the past few decades, BMCI algorithms have been developed, including methods based on subspaces (SS) [1, 2], methods based on cross-correlation of the second-order statistics [3, 4], and least squares (LS) [5, 6] and maximum likelihood (ML) methods [7]. The most advanced approaches are those derived from the normalized multichannel frequency-domain least mean square (NMCFLMS) algorithm [8–12]. Compared with the earlier time-domain adaptive algorithm [13], the classical NMCFLMS algorithm [8] leverages the fast Fourier transform (FFT) to adaptively estimate the impulse responses of single-input multiple-output (SIMO) systems in the frequency-domain, rendering it computationally efficient and particularly well-suited for practical applications. However, most existing BMCI algorithms have been evaluated primarily in conventional communication-oriented simulations. Their behavior in acoustic scenarios remains less well understood, where speech excitation, background noise, reverberation, and unknown channel lengths jointly affect identification accuracy.

BMCI has also been incorporated into acoustic applications that rely on acoustic transfer functions (ATFs), such as time-delay estimation, source localization, multichannel equalization, and dereverberation. For time delay estimation (TDE), He et al. [14] leveraged an M-estimator to define a robust cost function for BMCI algorithms to achieve acoustic source localization. In the applications of multichannel equalization, Guidorzi et al. [15] proposed to jointly estimate channel parameters, noise variances, and model order via a weighted least-squares approach, enabling effective equalization under non-stationary noise environments. In the context of multichannel equalization (or inverse filtering), Li et al. [16] addressed the BMCI problem in the short-time Fourier transform (STFT) domain by formulating it as the identification of convolutional transfer functions (CTFs), and adopted nonnegative multichannel equalization to tackle the multichannel dereverberation problem. Recently, Chen et al. [17] first used a linear MMSE estimator to estimate CTF coefficients, and the transformed ATFs are used for speech dereverberation via inverse filtering. However, CTFs provide an STFT-domain approximation to time-domain convolution. When the CTFs are estimated from critically sampled STFT representations, as in [16], the resulting inverse filtering may not exactly satisfy the exact multiple-input/output inverse theorem (MINT) [18] conditions associated with perfect reconstruction.

This paper provides a unified evaluation of existing BMCI algorithms in acoustic scenarios, focusing on both multichannel identification accuracy and their applicability in MINT-based inverse filtering. The contributions are twofold. First, to the best of our knowledge, this is the first study to fairly compare representative BMCI algorithms under simulated and measured acoustic conditions and to analyze their robustness to filter-length variations. Second, we integrate NMCFLMS-class adaptive algorithms with MINT for multichannel speech dereverberation, using the estimated RIRs as prior knowledge for inverse filtering. The experimental results show that NMCFLMS-class methods are better suited than solution-based BMCI algorithms for real acoustic channel estimation, and that the proposed framework achieves higher PESQ than weighted prediction er-

ror (WPE) and generalized WPE dereverberation methods [19, 20] across various environments, even under low-SNR conditions.

2 Background

This section briefly reviews BMCI algorithms and their connection to MINT-based speech dereverberation. Detailed formulations are given in Section 3.

2.1 BMCI Algorithms

The BMCI problem is commonly formulated as blind identification of a single-input multiple-output (SIMO) system, where the source signal is unknown and the channel responses are inferred from multichannel microphone observations. The system is generally assumed to satisfy two identifiability conditions [8]: 1) the polynomials formed by the RIRs of different channels are coprime, i.e., the channel impulse responses do not share common zeros; 2) the autocorrelation matrix of the source signal is full rank. Traditional BMCI algorithms mainly exploit second-order statistics (SOS) and inter-channel cross-relations. Subspace-based BMCI algorithms seek analytical solutions, whereas least-squares (LS) [6] and maximum-likelihood (ML) methods [7] are typically formulated as optimization problems. These algorithms can be computationally expensive for long time-domain impulse responses. To address this issue, Huang et al. [8] proposed the normalized multichannel frequency-domain LMS (NMCFLMS) algorithm for BMCI, where block frequency-domain processing reduces the cost of long convolutions and forms the basis for later frequency-domain BMCI algorithms. Moreover, Qiu et al. [21] recently proposed an NKP-based NMCFLMS for higher computational efficiency, which has a faster convergence rate and faster tracking than the traditional NMCFLMS algorithm. However, the NMCFLMS-class algorithms often face challenges in noisy environments. For better robustness under noisy conditions, many studies refined this family through several noise-robust NMCFLMS (RNMCFMLS) algorithms [10–12].

Among these BMCI algorithms, NMCFLMS was successfully applied to speech separation and dereverberation tasks [22]. Conversely, speech processing techniques can also be used to facilitate acoustic channel identification. A common strategy in recent studies is to use a “pseudo-reference signal” to alleviate the ill-posedness of blind estimation. For example, Meyer-Kahlen et al. [23] estimated directional RIRs via relative transfer function (RTF) identification between beamformer outputs and spherical harmonics. Taking this inspiration, Deppisch et al. [24] proposed generating a pseudo-reference signal via dereverberation and beamforming, followed by frequency-domain multichannel Wiener filtering, for spatial RIR estimation from smart glasses. Similarly, Chen et al. [17] exploited WPE dereverberation, delay-and-sum (DAS) beamforming, and source localization to estimate CTFs in the STFT domain and reconstruct ATFs. The resulting ATFs can then be used directly for speech dereverberation, as described in Section 2.2.

With the development of deep learning (DL), several DL-based RIR estimation methods have also been proposed [25–29]. However, most of them focus on single-channel RIR estimation, while BMCI algorithms largely remain within the traditional signal-processing framework. Thus, these DL-based models do not explicitly exploit the cross-channel structure that encodes directional information in multichannel acoustic responses. Moreover, DL-based models often require extensive context-specific measured or simulated data for different microphone array configurations. Thus, most current BMCI algorithms were originally developed for communication environments, where the simulation signals are often modeled as stationary or white processes. In contrast, audio signals are colored, non-stationary, and time-varying. This difference makes BMCI in real acoustic environments substantially more challenging. Therefore, the first goal of this paper is to evaluate current BMCI algorithms under consistent conditions and to investigate their effectiveness in estimating the same acoustic systems.

2.2 MINT-based Speech Dereverberation

Once reliable multichannel RIR estimates are available, they can be exploited for speech dereverberation via inverse filtering. The dereverberation problem shares the same mathematical foundation as channel equalization in communications, namely deconvolution.

Theoretically, classical MINT can perfectly recover the source signal under ideal channel conditions. MINT computes inverse filters by forcing the cascade of the channels and inverse filters to approximate a delta function, after which the inverse filters are applied to the microphone signals. Nevertheless, MINT is highly sensitive to filter perturbations and additive noise. Another use of MINT is to validate the accuracy of the RIR estimation. For example, Chen et al. [17] initially used WPE and DAS approaches to extract the source signal, which is obtained by computing the time difference of arrival (TDOA). The preprocessed output was then used for CTF estimation through a Wiener or Kalman filter. Finally, MINT was employed to evaluate different BMCI algorithms, since MINT dereverberation relies on accurate ATF estimates.

Practical dereverberation methods such as WPE [19] avoid explicit channel inversion by estimating and subtracting late reverberation. However, this can lack the theoretical exactness promised by multichannel inverse filtering. Recent progress in BMCI and speech dereverberation provides a possibility to leverage credible channel estimates for effective inverse-filtering-based speech recovery, but MINT-based dereverberation remains sensitive to channel conditioning and noise amplification. Therefore, another goal of this work is to propose a robust MINT-based dereverberation method that combines MINT with BMCI algorithms.

3 Proposed Method

In this section, we present a dereverberation framework combining BMCI-based RIR estimates with MINT inverse filtering. Throughout this paper, RIRs refer to multichannel RIRs unless otherwise specified.

Given multichannel reverberant observations, the proposed method first estimates the acoustic channels using an NMCFLMS-class BMCI algorithm and then constructs inverse filters following the MINT principle. When the observations are corrupted by noise, spectral subtraction [30, 31] is incorporated as a preprocessing step to provide more reliable inputs for BMCI and improve perceptual quality after dereverberation. The overall pipeline is illustrated in Fig. 1. Core derivation of the NMCFLMS-class algorithms is first presented, followed by a description of the proposed BMCI-MINT dereverberation pipeline.

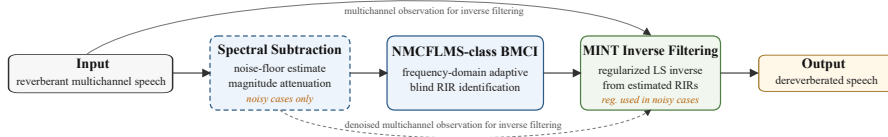


Fig. 1. Flowchart of the proposed BMCI-based MINT dereverberation framework.

3.1 NMCFLMS-class BMCI Algorithms

Following previous research, this paper considers a single-input multiple-output (SIMO) system with finite impulse response (FIR) channels, which satisfies the identifiability conditions described in Section 2.1.

Given an observed multichannel signal $\mathbf{x}(n)$ captured by a microphone array (consisting of M microphones) with multichannel impulse response $\mathbf{h}(n)$, the input-output relationship of the SIMO system can be expressed as:

$$\mathbf{x}(n) = \mathbf{h}(n) * s(n) = \sum_{l=0}^{L-1} \mathbf{h}(l)s(n-l), \quad (1)$$

where $*$ denotes linear convolution, L is the per-channel length of the RIR, and $s(n)$ is the source signal. In noisy environments, the additive noise signal $\mathbf{v}(n)$ is incorporated into Eq. (1), yielding the contaminated multichannel observation:

$$\mathbf{y}(n) = \mathbf{x}(n) + \mathbf{v}(n) = \sum_{l=0}^{L-1} \mathbf{h}(l)s(n-l) + \mathbf{v}(n). \quad (2)$$

Based on the above signal model, this subsection reviews several NMCFLMS-class BMCI algorithms, clarifying the RIR estimation process and tracing their evolution from the standard NMCFLMS to the variants used in our approach.

NMCFLMS Algorithm. Essentially, all NMCFLMS-class algorithms first compute a cost function in the frequency domain as follows:

$$\mathcal{J}_F(m) = \sum_{i=1}^{M-1} \sum_{j=i+1}^M \mathbf{e}_{ij}^H(m) \mathbf{e}_{ij}(m), \quad (3)$$

where $\mathbf{e}_{ij}(m)$ is the frequency-domain block error computed using FFT, determined by the cross-relation between two distinct acoustic channels (the i -th and the j -th channel). Because the presence of noise leads to biased RIR estimates, the corresponding cross-relation error is defined as:

$$\mathbf{e}_{ij}(m) = \mathbf{D}_{x_i}(m) \hat{\mathbf{h}}_j(m) - \mathbf{D}_{x_j}(m) \hat{\mathbf{h}}_i(m). \quad (4)$$

where $\mathbf{D}_{x_i}(m)$ is a diagonal matrix constructed from the FFT of the observed signal $x_i(n)$, and $\hat{\mathbf{h}}_i(m)$ denotes the frequency-domain estimate of the i -th channel RIR. When there is no noise and the estimation is accurate, the cross-relation error $\mathbf{e}_{ij}(m)$ tends to zero; otherwise, its magnitude reflects the estimation error.

By zero-padding the signal to length L , the spectrogram of the k -th channel in the m -th time block is obtained, which effectively reduces the truncation error inherent in the overlap-save technique. Subsequently, the NMCFLMS algorithm employs Newton's method [13] for the iterative update of the impulse response:

$$\hat{\mathbf{h}}_k^{10}(m+1) = \hat{\mathbf{h}}_k^{10}(m) - 2\rho \mathbf{W}_{2L \times 2L}^{10} \mathbf{P}_k^{-1}(m) \sum_{i=1}^M \mathbf{D}_{x_i}^*(m) \mathbf{e}_{ik}^{01}(m), \quad k = 1, 2, \dots, M \quad (5)$$

where $\hat{\mathbf{h}}_k^{10}(m)$ is the estimated RIR of the k -th channel at block m , $\rho \in (0, 2)$ is the step size, and $\mathbf{W}_{2L \times 2L}^{10} = 0.5 \mathbf{I}_{2L \times 2L}^{10}$ is the projection matrix. The term $\mathbf{P}_k^{-1}(m)$ is the inverse power matrix, and $\mathbf{D}_{x_i}^*(m)$ is the conjugate data matrix.

RNMCFLMS Algorithm. Under high-noise conditions, the performance of the NMCFLMS algorithm deteriorates significantly. To improve the noise robustness of BMCI algorithms, RNMCFLMS [10] introduces a logarithmic spectral penalty term into Eq. (3), which can be expressed as:

$$\mathcal{J}_R(m) = \mathcal{J}_F(m) - \beta(m) \sum_{i=1}^M \sum_{j=0}^{L-1} \log \left(\left| \hat{\mathbf{h}}_{i,j}(m) \right|^2 \right), \quad (6)$$

where $\beta(m)$ is an updating Lagrange multiplier. Based on Eq. (5), the update equation of the RNMCFLMS algorithm is:

$$\hat{\mathbf{h}}_k^{10}(m+1) = \hat{\mathbf{h}}_k^{10}(m) - \rho \mathbf{P}_k^{-1}(m) \sum_{i=1}^M \mathbf{D}_i^*(m) \mathbf{e}_{ik}^{01}(m) + \rho \beta(m) \nabla \mathcal{J}_k^{10}(m), \quad (7)$$

where $\nabla \mathcal{J}_k^{10}(m)$ is the spectral penalty gradient with respect to $\hat{\mathbf{h}}_k^{10}(m)$. By incorporating a logarithmic penalty term, the RNMCFLMS algorithm not only promotes sparsity in the impulse responses but also encourages spectral flatness, thereby preventing the estimated RIRs from converging to trivial solutions.

ℓ_p -RNMCFMLS Algorithm. In practice, the logarithmic spectral constraint in the RNMCFMLS algorithm is not well-suited for real acoustic RIR estimation, as RIRs are inherently sparse and their degree of sparsity varies across different acoustic environments. To address this issue, the ℓ_p -RNMCFMLS algorithm [11] replaces the logarithmic penalty from RNMCFMLS with an ℓ_p -norm sparsity constraint. The ℓ_p -RNMCFMLS algorithm introduces an ℓ_p -norm penalty term:

$$\mathcal{J}_{FP}(m) = \left\| \hat{\mathbf{h}}_k^{10}(m) \right\|_{\ell_p}^p, \quad (8)$$

where $1 < p \leq 2$ controls the degree of sparsity enforcement. Accordingly, the update equation of the ℓ_p -RNMCFMLS algorithm is formulated as:

$$\hat{\mathbf{h}}_k^{10}(m+1) = \hat{\mathbf{h}}_k^{10}(m) - \rho \nabla \mathcal{J}_{F,k}(m) + \rho \beta(m) \nabla \mathcal{J}_{FP,k}(m), \quad (9)$$

where $\nabla \mathcal{J}_{F,k}(m)$ denotes the gradient of Eq. (3), and $\nabla \mathcal{J}_{FP,k}(m)$ denotes the gradient of the ℓ_p -norm sparsity penalty defined in Eq. (8).

El_p -RNMCFMLS Algorithm. Previous NMCFLMS-class algorithms generally exploit only the magnitude information of the impulse responses, while neglecting the structural information of the phase spectrogram. However, the phase of RIRs becomes increasingly nonlinear as reverberation intensifies, which adversely affects the accuracy of channel estimation. To mitigate this effect, the El_p -RNMCFMLS algorithm [12] incorporates a phase-based penalty term that constrains the second-order derivative of the phase to be close to zero. The modified cost function is formulated as:

$$\mathcal{J}_E(m) = \mathcal{J}_F(m) - \eta(m) \mathcal{J}_{FP}(m) + \eta_G(m) \mathcal{J}_{FG}(m), \quad (10)$$

where $\mathcal{J}_{FG}(m)$ is the phase penalty term defined as the squared summation of the second-order numerical differentiation of the phase spectrogram, and $\eta_G(m)$ is the adaptive Lagrangian multiplier associated with this phase-related penalty term. The update equation of the El_p -RNMCFMLS algorithm is then given by:

$$\hat{\mathbf{h}}_k^{10}(m+1) = \hat{\mathbf{h}}_k^{10}(m) - \rho \nabla \mathcal{J}_{F,k}(m) + \rho \beta(m) \nabla \mathcal{J}_{FP,k}(m) - \rho \eta_G(m) \nabla \mathcal{J}_{FG,k}(m). \quad (11)$$

Furthermore, to improve the steady-state performance, the El_p -RNMCFMLS algorithm divides the estimated RIRs into significant and insignificant parts based on their magnitude relative to a predefined threshold. After a transition time t_0 determined by the convergence of the cost function, different p values are applied to the two parts ($p_1 = p$ for significant components and $p_2 = 1$ for insignificant components). This refinement enhances the steady-state identification accuracy without compromising the convergence speed.

3.2 MINT Inverse Filtering

Algorithm 1 Dereverberation Using NMCFLMS-class Algorithms and MINT

Require: Reverberant signal $\mathbf{x}[n] = [x_1[n], \dots, x_M[n]]^T$, BMCI algorithm $\mathcal{A}_{\text{BMCI}}$, filter length L , step size ρ , observation repeat count N_r , averaging length K_{avg} , inverse-filter length L_g , target delay τ , and regularization parameter λ .

Ensure: Dereverberated signal $\hat{s}[n]$.

- 1: Set block length $B = 2L$ and hop size $R = L$.
- 2: Repeat the reverberant signal N_r times to form $\mathbf{x}_{\text{BMCI}}[n]$ and determine the corresponding number of overlapping update blocks N_{blk} .
- 3: Initialize $\mathcal{A}_{\text{BMCI}}$ with M , L , ρ , and algorithm-specific parameters Θ_{BMCI} (e.g., the sparsity exponent p for Lp-RNMCFLMS and ELp-RNMCFLMS algorithms).
- 4: **for** $b = 1, \dots, N_{\text{blk}}$ **do**
- 5: Extract the b -th overlapping multichannel block $\mathbf{X}_b$ from $\mathbf{x}_{\text{BMCI}}[n]$ using block length B and hop size R .
- 6: Update the multichannel RIR estimates using $\mathcal{A}_{\text{BMCI}}$:

$$\{\hat{\mathbf{h}}_m^{(b)}\}_{m=1}^M = \mathcal{A}_{\text{BMCI}}(\mathbf{X}_b, \{\hat{\mathbf{h}}_m^{(b-1)}\}_{m=1}^M).$$

- 7: Normalize the stacked RIR estimates to unit norm and store $\{\hat{\mathbf{h}}_m^{(b)}\}_{m=1}^M$.
- 8: **end for**
- 9: Average the most recent K_{avg} estimates:

$$\bar{\mathbf{h}}_m = \frac{1}{K_{\text{avg}}} \sum_{b=N_{\text{blk}}-K_{\text{avg}}+1}^{N_{\text{blk}}} \hat{\mathbf{h}}_m^{(b)}, \quad m = 1, \dots, M.$$

- 10: Construct the multichannel convolution matrix

$$\mathbf{H} = [\mathbf{H}_1, \mathbf{H}_2, \dots, \mathbf{H}_M],$$

where $\mathbf{H}_m$ is formed from $\bar{\mathbf{h}}_m$.

- 11: Define the inverse filter vector $\mathbf{g}$ and the delayed target impulse vector $\mathbf{d}_\tau$.
- 12: Estimate the regularized MINT inverse filters:

$$\mathbf{g}^* = \arg \min_{\mathbf{g}} \|\mathbf{H}\mathbf{g} - \mathbf{d}_\tau\|_2^2 + \lambda \|\mathbf{g}\|_2^2.$$

- 13: Split optimal solution $\mathbf{g}^*$ into channel-wise filters $\{\mathbf{g}_m\}_{m=1}^M$.
- 14: Recover the source estimate:

$$\hat{s}[n] = \sum_{m=1}^M x_m[n] * g_m[n].$$

- 15: **return** $\hat{s}[n]$.
-

After the multichannel RIRs are blindly estimated by the above NMCFLMS-class BMCI algorithms, these estimates serve as the acoustic channel model for

MINT-based inverse filtering. In this way, all BMCI approaches share the same inverse-filtering stage, and their differences in final dereverberation performance are mainly attributable to the reliability of the estimated RIRs. For clarity, the detailed processing procedure is summarized in Algorithm 1. The procedure is applicable to all NMCFLMS-class BMCI algorithms just introduced.

Based on practical experience reported in [10, 12, 32], NMCFLMS-class algorithms generally require sufficiently long speech excitation (typically several minutes) to obtain converged estimates. However, such long recordings are not always available in real scenarios. In the proposed approach, the observed signal is repeated N_r times to form the BMCI input. This provides more update samples for the adaptive estimation process and enables more accurate RIR estimates, even when the available observation is only a few seconds long.

Since the RIR estimates used as prior knowledge for MINT are obtained from BMCI algorithms, their estimation errors may amplify noise perturbations or lead to unstable inverse filters in the exact MINT formulation [16]. For this reason, our implementation uses a Tikhonov-regularized least-squares formulation instead of enforcing the exact inverse. The regularization is adopted only in noisy cases to reduce the sensitivity of exact MINT to RIR estimation errors and ill-conditioned convolution matrices, thereby improving numerical robustness.

A spectral subtraction front-end [30] is also applied before BMCI. It performs channel-wise magnitude subtraction using a 10th-percentile noise-floor estimate, with a subtraction factor of 1.2 and a 2% floor relative to the original magnitude. Its effectiveness is empirically evaluated by comparing the untreated noisy and spectral-subtracted cases under different SNRs.

4 Experimental Setups

4.1 Baselines

The four NMCFLMS-based BMCI algorithms described in the previous section are used as iterative baselines, which are all representative adaptive iterative approaches. Meanwhile, four classical solution-based BMCI baselines are considered: the subspace (SS) method [2], the minimum noise subspace (MNS) method [33], the batch cross-relation (CR) method [6, 34], and the block-wise multichannel Newton (BMCN) algorithm [8, 13].

For dereverberation baselines, we select the WPE method [19] and the generalized WPE (GWPE) method [20] for comparison, both of which are state-of-the-art signal processing dereverberation techniques.

4.2 Data Simulation

BMCI Experiments: We evaluated the BMCI algorithms using one simulated acoustic impulse-response dataset and one measured dataset under comparable conditions. The simulated response modeled a car cabin (256 taps) using the small-room configuration from the study [8]. The measured responses were

captured in an anechoic chamber and a medium reverberant room using an 8-channel non-uniform linear microphone array; four channels ($M = 4$) with inter-microphone spacings of 160–100–160 mm were selected for blind identification. An artificial mouth was placed 1 m from the array at azimuths of 90° (front), 135° (diagonal-front), and 180° (side). At each position, we obtained reference impulse responses via exponential sweep. For BMCI, we used white Gaussian noise and eight speech utterances—four English (from TIMIT dataset [31]) and four Mandarin (from THCHS-30 dataset [35]). All utterances were 10 s long and sampled at 16 kHz. The filter length was set to $L = 1024$ samples to assess multichannel estimation accuracy.

Dereverberation Experiments: For the dereverberation task, two image-source-generated RIRs are used as the true acoustic systems under both quiet and noisy conditions. The first RIR is the same simulated small-room RIR as that used in the BMCI experiments and is regarded as the true RIR to be identified during the BMCI stage. The second RIR is generated using the simulation configuration adopted in the El_p -RNMCFMLS study [12], with a room size of $9 \times 7 \times 5$ m and a reverberation time of $T_{60} = 0.5$ s. The second RIR was truncated to 1024 taps to form finite-length FIR systems and to assess how BMCI accuracy affects MINT-based dereverberation. The utterance “MAHH0_SI1294” from the TIMIT dataset was selected as the common excitation signal.

4.3 Implementations

Four NMCFLMS-class BMCI algorithms were implemented with identical filter initialization, step-size ρ ($= 0.5$), and normalization setups using the same evaluation setting following [12]. The ℓ_p -RNMCFMLS and El_p -RNMCFMLS algorithms also adopt a set of p values (1.0, 1.3, and 1.6) following the original paper [12]. In the most recent BMCI studies [11, 12, 32], the normalized projection misalignment (NPM) was used to quantify the RIR estimation accuracy, and a lower NPM indicates better BMCI performance. But swept-sine measurement and blind estimation suffer from different source distortions and inherent scale ambiguity, making accurate waveform comparison between the measurements and estimates difficult. To this end, we compute NPM based on relative impulse response [36] for measured channels. For each non-reference microphone, the relative transfer function (RTF) is computed with respect to the first channel, converted back to the time-domain relative impulse response, and then compared with the ground truth using the same projection-misalignment criterion as NPM. This metric is defined as ReIR NPM, which is directly relevant to multichannel inverse filtering, emphasizing the accuracy of inter-channel relative responses.

In the dereverberation task, the quality of the dereverberated speech signal is evaluated using PESQ [37] when MINT is applied with the estimated RIRs. To investigate the applicability of the proposed framework in noisy environments, uncorrelated WGN is added to the reverberant signals at SNRs of 10 dB, 20 dB, and 30 dB after convolution of the source with the RIRs. For Algorithm 1, the filter/inverse-filter length is equal to the RIR length L . The hyperparameters ρ , K_{Avg} , τ , and λ are set to 0.1, 100, $L - 1$, and 0.1, respectively.

5 Results

5.1 Multichannel System Identification

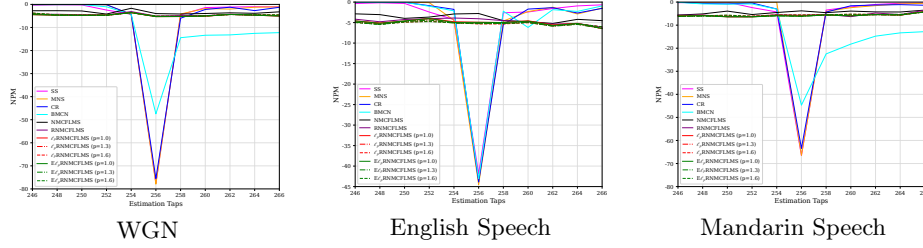


Fig. 2. NPM results of different BMCI algorithms under varying estimation lengths.

Tap Tolerance Analysis in the Simulation Environment Since previous methods require a preset filter length, we conduct a tolerance analysis using the simulated 256-tap RIR with a WGN sequence, a TIMIT English utterance and a THCHS-30 Mandarin utterance. When the assumed tap length deviates from the true RIR length, conventional solution-based BMCI algorithms suffer severe performance degradation. As shown in Fig. 2, SS, MNS, CR, and BMCN achieve their best NPM at 256 taps, but degrade sharply under length mismatch. When speech signals are used as excitation, the performance of each algorithm decreases significantly compared to the WGN case. Because the true RIR length is generally unknown in practice, conventional solution-based methods are ill-suited for acoustic system identification. By contrast, NMCFLMS-class algorithms remain stable under tap-length mismatch, despite not achieving the lowest NPM at 256 taps, and thus show greater application potential. Although BMCN shows robustness similar to that of NMCFLMS-class algorithms under overestimated setups (estimation taps $\geq$ RIR taps), its high computational cost limits practicability, as will be discussed later.

BMCI Comparison in the Measured Acoustic Environment. This experiment considers anechoic and medium reverberant conditions, representing easy and realistic acoustic environments. The results are shown in Fig. 3, where the averaged ReIR NPMs of speech samples are presented. In line with the preceding tolerance analysis, the NMCFLMS-class BMCI algorithms achieve lower ReIR NPM values than the solution-based BMCI methods in the both rooms, indicating more robust and accurate estimation of the measured acoustic impulse responses. BMCN and NMCFLMS-class methods are equally effective under anechoic conditions, but BMCN fails in the reverberant room while incurring much higher computational cost. For identifying an impulse response with $L = 1024$, BMCN took about 6–9 hours per estimation, whereas NMCFLMS-class methods completed the estimates within seconds to minutes.

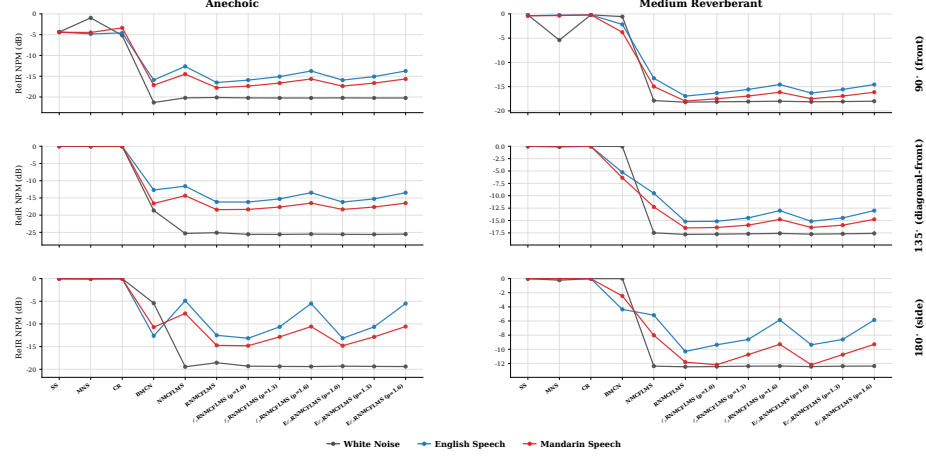


Fig. 3. ReIR NPM of estimates obtained using WGN, English speech, and Mandarin speech as excitation signals in the anechoic and medium reverberant environments.

5.2 BMCI-based RIR Estimates for Dereverberation

To comprehensively evaluate the proposed framework, this subsection considers one quiet case and noisy cases with SNRs ranging from 5 to 30 dB. Hereafter, $p = 1$ is used for both ℓ_p -RNMCFMLS and El_p -RNMCFMLS, as it yielded the best overall performance among the tested p values. In addition, the ρ values of BMCI algorithms are set to 0.1 for stable convergence.

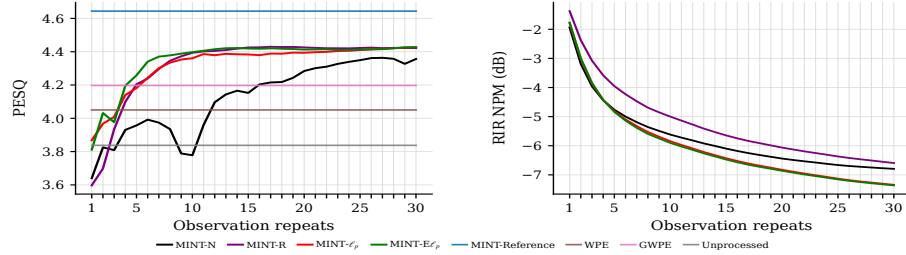


Fig. 4. PESQ and BMCI NPM results over repetition count N_r in steps of 1 for a 256-tap RIR excited by TIMIT speech under the noise-free condition. Here, MINT- N / R / ℓ_p / El_p refer to MINT using RIRs estimated by NMCFLMS-class algorithms.

Results under Noise-free Condition. Fig. 4 presents the PESQ scores of the evaluated dereverberation methods across different numbers of observation repetitions, together with the corresponding NPM values of the BMCI algorithms.

Table 1. PESQ comparison across different methods ($T_{60} = 0.5$ s, $L = 1024$).

	Unprocessed	WPE	GWPE	MINT-N	MINT-R	MINT- ℓ_p	MINT- E_{ℓ_p}	MINT-Reference
Without spectral subtraction front-end								
PESQ (SNR=30 dB, $\lambda = 0.1$)	1.8370	2.3508	2.6416	2.7236	2.8683	2.8853	2.7970	3.0604
PESQ (SNR=20 dB, $\lambda = 0.1$)	1.6112	1.7441	1.8438	2.2578	2.2741	2.2361	2.1665	<u>2.2670</u>
PESQ (SNR=10 dB, $\lambda = 0.1$)	1.2647	1.2577	1.2794	1.6787	1.3981	1.3908	1.3808	<u>1.4844</u>
With spectral subtraction front-end								
PESQ (SNR=30 dB, $\lambda = 0.1$)	1.9014	2.3239	2.8025	2.8357	<u>2.9864</u>	2.9162	2.8740	3.1885
PESQ (SNR=20 dB, $\lambda = 0.1$)	1.7158	1.8525	2.0461	2.3550	2.4142	<u>2.4799</u>	2.4582	2.5522
PESQ (SNR=10 dB, $\lambda = 0.1$)	1.3673	1.3548	1.3976	1.7940	1.5357	1.5349	1.5389	<u>1.6440</u>
PESQ (SNR=5 dB, $\lambda = 0.1$)	1.2275	1.1960	1.2049	1.6328	1.2797	1.2752	1.2708	<u>1.3643</u>
PESQ (SNR=5 dB, $\lambda = 0.01$)	1.2275	1.1960	1.2049	1.6218	1.2507	1.2188	1.2157	<u>1.2529</u>
PESQ (SNR=5 dB, $\lambda = 0.001$)	<u>1.2275</u>	1.1960	1.2049	1.4666	1.2171	1.1788	1.1797	1.1886
PESQ (SNR=5 dB, $\lambda = 0$)	1.2275	1.1960	1.2049	<u>1.2095</u>	1.0755	1.0885	1.1450	1.1593

MINT-Reference uses the reference RIR as prior information for MINT-based inverse filtering and achieves nearly perfect reconstruction under the noise-free condition (PESQ over 4.6). WPE and GWPE provide limited improvement in this short-RIR case, while BMCI-MINT methods approach MINT-Reference as the repetitions increase. MINT-R sometimes achieves the highest PESQ score, despite its poorer NPM than other BMCI algorithms, suggesting that NPM may not fully reflect perceptual quality of the reconstruction. Hereafter, we set $N_r = 30$ for the BMCI-based MINT dereverberation methods, as their performance has nearly saturated at 30 repetitions.

Results under White Noise Conditions. To evaluate the proposed pipeline under stronger reverberation and noisy conditions, we conduct a robustness analysis across different SNR levels. For fairness, the spectral subtraction front-end [30] is applied for all methods, and the results are reported in Table 1. Since noisy and pre-denoised observations no longer exactly satisfy the original linear convolution model, MINT-Reference is not a strict PESQ upper bound. In some SNR cases, BMCI estimates obtained from the pre-denoised signals may better match the effective channel response and yield higher PESQ. Overall, BMCI-based MINT dereverberation outperforms WPE and GWPE in PESQ, whether at low SNRs or not. Finally, different λ values are tested for MINT at 5 dB SNR to verify the regularization setting under noisy cases. However, the high PESQ of MINT-N does not truly reflect its dereverberation performance, as the unreliable RIR estimates cause the regularized inverse filters to softly suppress more noise components without adequately eliminating reverberation.

6 Conclusion

In this paper, we compare existing BMCI algorithms and evaluate their feasibility for MINT-based speech dereverberation from two perspectives: acoustic channel estimation accuracy and MINT-based dereverberation performance. Solution-based BMCI algorithms perform well when the filter length is correctly specified, but fail under length mismatch. In contrast, NMCFLMS-class algorithms are more robust for acoustic BMCI and can provide useful RIR priors for MINT-based inverse filtering. The proposed pipeline outperforms the WPE and GWPE methods in the dereverberation task across noise-free and low-SNR cases.

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